

IAN WAN

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EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Systems Engineering and Design
Industrial & Enterprise Systems Engineering Dept.

Expected Graduation May 2029

GPA: 3.98/4.00

- **Relevant Coursework:** Business Side of Engineering, Engineering Graphics & Design, Intro to Milling for Engineers, Intro to Turning for Engineers, TIG Welding for Engineers

EXPERIENCE

Research Intern under Professor Emeritus Max F. Platzer

Jun. 2024 – Present

AeroHydro Research & Technology Associates

Fremont, CA

- Developing a bird-friendly, lower noise wind power generator that harvests energy by oscillating wings rather than spinning rotor blades, reducing by at least 25% the minimum wind speed for sustained oscillation
- Modeled the oscillator mechanism and its 3D-printed bearing mounts, centering rings, couplers, and end caps in Fusion 360, iterating the assembly with the faculty advisor before fabrication, then migrated the model to Creo Parametric for version control and assembly modification
- Built the primary support mount as a single rigid wing-and-shaft unit carried on two spaced bearings, constraining the wing to a single rotational degree of freedom and isolating the intended motion from off-axis loading
- Designed the machine's mechanical travel stop in Creo Parametric, winning adoption over the advisor's competing concept on packaging and efficiency grounds, and slotting it so travel limits adjust without fabricating new parts
- Designed and fabricated an open-return wind tunnel against 7 written design constraints and a \$500 cap, producing a controlled test bed
- Ran the program's first closed-loop field campaign at a working wind farm, executing a 7-run sweep of the controller's two switching parameters, obtaining sustained oscillation at every trial
- Specified and drove a two-revision firmware reduction from a two-loop, four-gain controller to a switching scheme with none, trading system-specific tailoring for field debuggability
- Diagnosed a multi-week stall as a setting left too low rather than broken hardware, restoring motion with no new parts
- Tracked project budget and sourced structural and electronic components across 6 categories, managing roughly \$1,000 per year of build spend and reconciling purchases with the advisor
- Negotiated a joint test matrix with a sister research group at the University of Brighton, and itemized the 5 measurements their wind tunnel would return so field data could be validated against controlled conditions
- Ran recurring technical reviews with the faculty advisor and collaborators, maintaining a 101-entry decision log and 20-item open-risk register, and delivered the final field campaign against a fixed return-to-school deadline

PROJECTS

Steering Wheel Project Lead

Aug. 2025 – Present

Illini Electric Motorsports, Formula SAE Electric; Project Lead since July 2026

Champaign, IL

- Own the steering wheel assembly on the 2026–27 car under the cockpit subsystem lead, covering requirements, scope, design reviews, the drawing package, and the build plan
- Baselined the outgoing wheel on mass, part count, bill-of-materials cost, driver egress time, and driver feedback before opening the redesign, so every later change is measured against a recorded starting point
- Produced 2D part drawings to ASME Y14.5 GD&T plus manufacturing-process documentation for the competition cost event, delivering machinist-ready drawings across the cockpit subsystem
- Machined jig components on a Bridgeport manual mill from own drawings, carrying the design from machinist-ready print through to finished hardware
- Designed a waterjet-ready U-joint thrust-force test jig in Creo Parametric, letting future project leads size the steering column without repeating physical testing

Team Pulsaris Research and Development Lead

Sep. 2022 – Mar. 2024

Conrad Challenge, Innovation and Entrepreneurship Competition

Fremont, CA

- Coordinated the R&D roadmap with a 2-person Business & Marketing division so technical work mapped to the venture's target customers and business model
- Redesigned the Apollo Lunar Module landing-gear down-lock, an 8-component cam-and-spring latch, as a single-piece 3D-printable compliant mechanism across 7 Fusion 360 revisions, cutting part count from 8 to 1
- Cut peak von Mises stress and peak displacement roughly $2\times$ each under a 25,000 N design load derived from Apollo strut test data, validated by linear static FEA
- Printed functional hardware prototypes at major revisions to test fit, range of motion, and latching
- Interfaced across the team's three-division structure to assemble the combined Innovation Brief within the competition's strict word and format limits, advancing the team to the Conrad "Innovator" distinction

Applications of Compliant Mechanisms Researcher

Feb. 2024 – Mar. 2024

Alameda County (ACSEF) and California State (CSEF) Science Fairs

Fremont, CA

- Benchmarked compliant against rigid designs on 3 test articles, raising minimum factor of safety $5.3\times$ (0.525 to 2.828) on a two-hole puncher mechanism at a matched 20 N load
- Load-tested TPU and PLA prototypes against a rigid baseline, comparing print material and process, the honeycomb holding over $90\times$ its own weight
- Ran a one-sample t -test on punched hole diameter, confirming the compliant puncher held the 7 mm commercial standard at the 0.05 significance level

SKILLS

Project Coordination: requirements and scope definition, schedule and budget tracking, cross-team and cross-functional coordination, drawing-package and documentation ownership, recurring design reviews, decision and open-risk logs

3D CAD and Design: Creo Parametric, Autodesk Fusion 360, machine and mechanism design, rapid prototyping

Manufacturing: GD&T (ASME Y14.5), design for manufacturing, part and assembly drawings, machinist-ready drawings, process documentation, Bridgeport manual mill, turning, TIG welding, waterjet design, design for additive manufacturing

Test and Measurement: test fixture and rig design, test planning, wind tunnel design and operation, Design of Experiments, hypothesis testing, failure-mode analysis, data logging

Analysis: Ansys Mechanical FEA, linear static stress analysis, MATLAB, Python, C++

Prototyping and Instrumentation: breadboarding, sensor and actuator integration, closed-loop control (Arduino)